Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 15)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain why the potential to increase food supplies by cultivating new land is limited.

[2 marks]

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0 1 . 2 Explain why importing food may reduce the availability of food in the area where it is produced.

[2 marks]

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0 1 . 3 Explain why organic agriculture is more than simply 'not using pesticides and artificial fertilisers'.

[2 marks]

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0 1 . 4 Evaluate the strategies that could be used to make agriculture more sustainable.

[9 marks]

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0 2 . 1 Which of the following best describes a **representative** sample? Tick one box.

[1 mark]

A A sample taken from the most convenient location ☐

B A sub-sample that accurately reflects the complete data set ☐

C The largest sample that can be collected ☐

D A sample that supports the hypothesis ☐

0 2 . 2 Describe how random sampling sites would be selected in an irregularly shaped meadow.

[2 marks]

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0 2 . 3 Explain why deliberately choosing sampling sites introduces bias.

[2 marks]

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0 3 . 1 Explain the difference between in-situ and ex-situ conservation and state when ex-situ conservation is needed.

[2 marks]

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0 3 . 2 Give **four** criteria used to select species for captive breeding and release programmes.

[4 marks]

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0 3 . 3 Explain why each of the following species would be difficult to keep or breed in captivity.

[4 marks]

Blue whale .....

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Large blue butterfly .....

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0 4 . 1 Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species ☐

B Mixed indigenous species with a mixed age structure ☐

C Clear-felling on a 40-year rotation ☐

D Removal of dead wood ☐

0 4 . 2 Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

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0 5 Table 1 shows the numbers of breeding pairs of two seabird species on an island before and after a rat eradication programme.

Table 1

Species	Breeding pairs before eradication	Breeding pairs 8 years after eradication
Manx shearwater	45	310
Storm petrel	12	140

0 5 . 1 Explain why oceanic islands have many endemic species but low overall biodiversity.

[4 marks]

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0 5 . 2 Use Table 1 to explain why introduced rats had such a severe effect on these birds.

[3 marks]

0 5 . 3 Explain how avian malaria threatens Hawaiian birds and how climate change worsens the threat.

[2 marks]

0 5 . 4 Give **one** other conservation measure for oceanic islands besides eradicating introduced species.

[1 mark]

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0 6 Table 1 compares two exploited species.

Table 1

Feature	Anchovy	Blue whale
Age at sexual maturity	1 year	10 years
Offspring per female per year	Thousands of eggs	1 calf every 2-3 years
Lifespan	3-4 years	80+ years
Parental care	None	Extensive

0 6 . 1 Use Table 1 to identify which species is r-selected and which is k-selected, and explain the difference between the strategies.

[3 marks]

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0 6 . 2 Explain why k-selected species are much more easily over-exploited than r-selected species.

[3 marks]

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0 6 . 3 State the equation used to forecast a change in population size.

[1 mark]

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0 6 . 4 Explain what is meant by **maximum sustainable yield** and how it relates to carrying capacity.

[3 marks]

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0 7 . 1 Explain what a carbon footprint should include when calculated for a car.

[3 marks]

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0 7 . 2 Describe what the WWF Living Planet Index measures and how it is used.

[2 marks]

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0 7 . 3 Describe how the Ol Pejeta reserve in Kenya combines commercial cattle ranching with wildlife conservation.

[5 marks]

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0 8 . 1 For each of the following forms of direct exploitation, give a named species whose population has been reduced.

[5 marks]

Food .....

Fashion .....

Entertainment / pets .....

Furniture and ornaments .....

Traditional medicines .....

0 8 . 2 Explain why species are deliberately eradicated. Give **three** categories with examples.

[3 marks]

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0 8 . 3 Explain why legal protection alone rarely prevents the decline of a species exploited for traditional medicine.

[2 marks]

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0 9 . 1 Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures ☐

B Reliance on non-renewable energy ☐

C Waste products that are the raw materials for other processes ☐

D Dispersal of elements so they cannot be re-used ☐

0 9 . 2 Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

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0 9 . 3 Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

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1 0 Table 1 shows the number of adult great crested newts caught in bottle traps in a pond over five years, and the number of ponds in the area in which the species was detected using environmental DNA (eDNA).

Table 1

Year	Adults caught in traps	Ponds with newts detected by eDNA (out of 20)
2019	14	6
2020	22	9
2021	18	11
2022	31	13
2023	27	15

1 0 . 1 Calculate the percentage increase in the number of ponds where newts were detected between 2019 and 2023.

[2 marks]

Answer ..... %

1 0 . 2 Explain how eDNA detects the presence of newts without catching them.

[2 marks]

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1 0 . 3 Suggest why trap catches are a less reliable measure of population change than the eDNA survey.

[2 marks]

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1 0 . 4 Evaluate the contribution of new technologies to ecological monitoring and wildlife conservation.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.

[25 marks]

OR

1 1 . 2 Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS